REVIEWER FOR MODULE 8 9 11 AND 12

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MODULE 8: NETWORK LAYER

8.1 NETWORK LAYER CHARACTERISTICS

1. **THE NETWORK LAYER**
- Provides services to allow end devices to exchange data
- IPv4 and IPv6 are the principle network layer communication protocols
- It has 4 basic operations:
    - **Addressing end devices**
    - **Encapsulation**
    - **Routing**
    - **De-encapsulation**

2. **IP ENCAPSULATION**
- IP encapsulates the transport layer segment
- IP can use either an IPv4 or IPv6 packet and not impact the layer 4 segment
- IP packet will be examined by all layer 3 devices as it traverses the network
- IP addressing does not change from source to destination

3. **CHARACTERISTICS OF IP**
- IP is meant to have low overhead and may be described as
    - **Connectionless**
        - IP does not establish a connection with the destination before sending the packet
        - There is no control information needed (synchronization, acknowledgements)
        - The destination will receive the packet when it arrives but no pre-notifications are sent by IP
        - If there is a need for connection-oriented traffic, then another protocol will handle this (typically TCP at the transport layer)
    - **Best Effort**
        - IP will not guarantee delivery of the packet
        - Ip has reduces overhead since there is no mechanism to resend data that is not received
        - IP does not expect acknowledgements
        - IP does not know if the other devices is operational of If it received the packet
    - **Media Independent**
        - **IP is unreliable:**
            - It cannot manage or fix undelivered or corrupt packets
            - IP cannot retransmit after an error
            - IP cannot realign out of sequence packets
            - IP must rely on other protocols for these functions
        - **IP is media Independent:**
            - IP does not concern itself with the type of frame required at the data link layer or the media type at the physical layer
            - IP can be sent over any media type:
                - COPPER
                - FIBER
                - WIRELESS
        - The network layer will establish the Maximum Transmission Unit (MTU)
            - Network layer receives this from control information sent by the data link layer
            - The network then establishes the MTU size
        - Fragmentation is when Layer 3 splits the IPv4 packet into smaller units
            - Fragmenting causes latency
            - IPv6 does not fragment packets
            - Example: Router goes from Ethernet to a slow WAN with a smaller MTU

8.2 IPv4 Packet
- **IPv4 Packet Header**
    - IPv4 is the primary communication protocol for the network layer
    - **Network header has many purposes:**
        - It ensures the packet is sent in the correct destination
        - It contains information for network layer processing in various fields
        - The information in the header is used by all layer 3 devices that handle the packet

- **IPv4 Packet Header Fields**
  - **The IPv4 network header characteristics:**
    - It is binary
    - Contains several fields of information
    - Diagram is read from left to right, 4 bytes per line
    - The two most important fields are the source and destination
  - **Significant Fields in the IPv4 Header:**
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Function	Description
Version	This will be for v4, as opposed to v6, a 4 bit field= 0100
Differentiated Services	Used for QoS: DiffServ – DS field or the older IntServ – ToS or Type of Service
Header Checksum	Detect corruption in the IPv4 header
Time to Live (TTL)	Layer 3 hop count. When it becomes zero the router will discard the packet.
Protocol	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Source IPv4 Address	32 bit source address
Destination IPV4 Address	32 bit destination address

8.3 IPv6 Packets

- **LIMITATIONS OF IPv4**
  - **IPv4 address depletion** - We have basically run out of IPv4 addressing
  - **Lack of end-to-end connectivity** - To make IPv4 survive the long, private addressing and NAT were created. This ended direct communication with public addressing
  - **Increased network complexity** - NAT was meant as temporary solution and creates issues on the network as a side effect of manipulating the network headers addressing. NAT causes latency and troubleshooting issues.

- **IPv6 OVERVIEW**
  - IPv6 was developed by Internet Engineering Task Force (IETF).
  - IPv6 overcomes the limitations of IPv4
  - **Improvements that IPv6 provides:**
    - **Increased address space** - based on 128 bit address, not 32 bits
    - **Improved packet handling** - simplified header with fewer fields
    - **Eliminates the need for NAT** - since there is a huge amount of addressing, there is no need to use private addressing internally and be mapped to a shared public address

- **IPv4 PACKET HEADER FIELDS IN THE IPv6 PACKET HEADER**
  - The IPv6 header is simplified, but not smaller.
  - The header is fixed at 40 Bytes or octets long.
  - Several IPv4 fields were removed to improve performance.
  - Some IPv4 fields were removed to improve performance:
    - Flag
    - Fragment Offset
    - Header Checksum

- **IPv6 PACKET HEADER**
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Function	Description
Version	This will be for v6, as opposed to v4, a 4 bit field= 0110
Traffic Class	Used for QoS: Equivalent to DiffServ – DS field
Flow Label	Informs device to handle identical flow labels the same way, 20 bit field
Payload Length	This 16-bit field indicates the length of the data portion or payload of the IPv6 packet
Next Header	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Hop Limit	Replaces TTL field Layer 3 hop count
Source IPv4 Address	128 bit source address
Destination IPV4 Address	128 bit destination address

- **IPv6 PACKET HEADER (CONT.)**

- IPv6 packet may also contain extension headers (EH).
- EH headers characteristics:
  - provide optional network layer information
  - are optional
  - are placed between IPv6 header and the payload
  - may be used for fragmentation, security, mobility support, etc
  - **Note:** Unlike IPv4, routers do not fragment IPv6 packets

8.4 HOW A HOST ROUTES

- **HOST FORWARDING DECISION**
  - Packets are always created at the source
  - Each host devices creates their own routing table.
  - A host can send packets to the following:
    - **Itself** – 127.0.0.1 (IPv4), ::1 (IPv6)
    - **Local Hosts** – destination is on the same LAN
    - **Remote Hosts** – devices are not on the same LAN

- **HOST FORWARDING DECISION (CONT.)**
  - The Source device determines whether the destination is local or remote
  - Method of determination:
    - IPv4 –Source uses its own IP address and Subnet mask, along with the destination IP address
    - IPv6 –Source uses the network address and prefix advertised by the local router
  - Local traffic is dumped out the host interface to be handled by an intermediary device.
  - Remote traffic is forwarded directly to the default gateway on the LAN

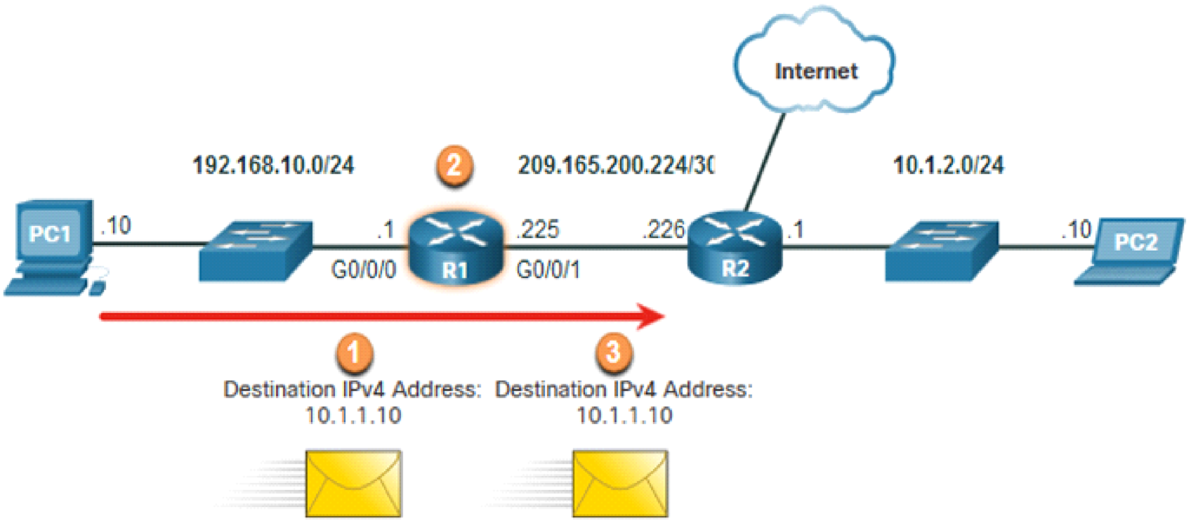
- **DEFAULT GATEWAY**
  - A router or layer 3 switch can be a default-gateway
  - Features of a default gateway (DGW):
    - It must have an IP address in the same range as the rest of the LAN.
    - It can accept data from the LAN and is capable of forwarding traffic off of the LAN.
    - It can route to other networks.
  - If a device has no default gateway or a bad default gateway, its traffic will not be able to leave the LAN

- **A HOST ROUTES TO THE DEFAULT GATEWAY**
  - The host will know the default gateway (DGW) either statically or through DHCP in IPv4.
  - IPv6 sends the DGW through a router solicitation (RS) or can be configured manually.
  - A DGW is static route which will be a last resort route in the routing table.
  - All device on the LAN will need the DGW of the router if they intend to send traffic remotely

- **HOST ROUTING TABLES**
  - On Windows, route print or netstat-r to display the PC routing table
  - Three sections displayed by these two commands:
    - Interface List – all potential interfaces and MAC addressing
    - IPv4 Routing Table
    - IPv6 Routing Table

8.5 INTRODUCTION TO ROUTING

- **ROUTER PACKET FORWARDING DECISION**

- 

  - Packet arrives on the Gigabit Ethernet 0/0/0 interface of router R1. R1 de-encapsulates the Layer 2 Ethernet header and trailer.
  - Router R1 examines the destination IPv4 address of the packet and searches for the best match in its IPv4 routing table. The route entry indicates that this packet is to be forwarded to router R2.
  - Router R1 encapsulates the packet into a new Ethernet header and trailer, and forwards the packet to the next hop router R2

- **IP ROUTER ROUTING TABLE**
  - **Directly Connected** – These routes are automatically added by the router, provided the interface is active and has addressing.
  - **Remote**–These are the routes the router does not have a direct connection and may be learned:
    - **Manually** – with a static route
    - **Dynamically** – by using a routing protocol to have the routers share their information with each other
  - **Default Route** – this forwards all traffic to a specific direction when there is not a match in the routing table

- **STATIC ROUTING**
  - Static Route Characteristics:
    - Must be configured manually
    - Must be adjusted manually by the administrator when there is a change in the topology
    - Good for small non-redundant networks
    - Often used in conjunction with a dynamic routing protocol for configuring a default route

- **DYNAMIC ROUTING**
  - Dynamic Routing:
    - Discover remote networks
    - Maintain up-to-date information
    - Choose the best path to the destination
    - Find new best paths when there is a topology change
    - Dynamic routing can also share static default routes with the other routers.

MODULE 9: ADDRESS RESOLUTION

9.1 MAC AND IP

DESTINATION ON SAME NETWORK

- There are two primary addresses assigned to a device on an Ethernet LAN:
  - **Layer 2 physical address (the MAC address)**– Used for NIC to NIC communications on the same Ethernet network.
  - **Layer 3 logical address (the IP address)**– Used to send the packet from the source device to the destination device.
- Layer 2 addresses are used to deliver frames from one NIC to another NIC on the same network. If a destination IP address is on the same network, the destination MAC address will be that of the destination device

DESTINATION ON REMOTE NETWORK

- When the destination IP address is on a remote network, the destination MAC address is that of the default gateway.
  - ARP is used by IPv4 to associate the IPv4 address of a device with the MAC address of the device NIC.
  - ICMPv6 is used by IPv6 to associate the IPv6 address of a device with the MAC address of the device NIC.

9.2 ARP

ARP OVERVIEW

- A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address.
- ARP provides two basic functions:
  - Resolving IPv4 addresses to MAC addresses
  - Maintaining an ARP table of IPv4 to MAC address mappings

ARP FUNCTIONS

- To send a frame, a device will search its ARP table for a destination IPv4 address and a corresponding MAC address.
- If the packet's destination IPv4 address is on the same network, the device will search the ARP table for the destination IPv4 address.
- If the destination IPv4 address is on a different network, the device will search the ARP table for the IPv4 address of the default gateway.
- If the device locates the IPv4 address, its corresponding MAC address is used as the destination MAC address in the frame.
- If there is no ARP table entry is found, then the device sends an ARP request

REMOVING ENTRIES FROM AN ARP TABLE

- Entries in the ARP table are not permanent and are removed when an ARP cache timer expires after a specified period of time.
- The duration of the ARP cache timer differs depending on the operating system. A
- ARP table entries can also be removed manually by the administrator.

ARP TABLES ON NETWORKING DEVICES

- The **show ip arp** command displays the ARP table on a Cisco router.
- The **arp –a** command displays the ARP table on a Windows 10 PC

ARP ISSUES - ARP BROADCATING AND ARP SPOOFING

- ARP requests are received and processed by every device on the local network
- Excessive ARP broadcasts can cause some reduction in performance.
- ARP replies can be spoofed by a threat actor to perform an ARP poisoning attack.
- Enterprise level switches include mitigation techniques to protect against ARP attacks

9.3 COPPER CABLING

IPv6 NEIGHBOR DISCOVERY MESSAGES

- IPv6 Neighbor Discovery (ND) protocol provides:
  - Address resolution
  - Router discovery
  - Redirection services
  - ICMPv6 Neighbor Solicitation (NS) and Neighbor Advertisement (NA) messages are used for device-to-device messaging such as address resolution.
  - ICMTPv6 Router Solicitation (RS) and Router Advertisement (RA) messages are used for messaging between devices and routers for router discovery.
  - ICMPv6 redirect messages are used by routers for better next-hop selection

IPv6 NEIGHBOR DISCOVERY - ADDRESS RESOLUTION

- IPv6 devices use ND to resolve the MAC address of a known IPv6 address.
- ICMPv6 Neighbor Solicitation messages are sent using special Ethernet and IPv6 multicast addresses.

MODULE 11: IPv4 ADDRESSING

11.1 IPv4 ADDRESS STRUCTURE

- **NETWORK AND HOST PORTIONS**
  - An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.
  - When determining the network portion versus the host portion, you must look at the 32-bit stream.
  - A subnet mask is used to determine the network and host portions.

- **THE SUBNET MASK**
  - To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.
  - The actual process used to identify the network and host portions is called ANDing.

- **THE PREFIX LENGTH**
  - A prefix length is a less cumbersome method used to identify a subnet mask address.

- The prefix length is the number of bits set to 1 in the subnet mask.
- It is written in "slash notation" therefore, count the number of bits in the subnet mask and prepend it with a slash

- **DETERMINING THE NETWORK: LOGICAL AND**
  - A logical AND Boolean operation is used in determining the network address.
  - Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
  - 1 AND 1 = 1, 0 AND 1 = 0, 1 AND 0 = 0, 0 AND 0 = 0
  - 1 = True and 0 = False
  - To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address

- **NETWORK, HOST, AND BROADCAST ADDRESSES**

	Network Portion			Host Portion	Host Bits
Subnet mask **255.255.255**.0 or **/24**	255 11111111	255 11111111	255 11111111	0 00000000	
Network address **192.168.10**.0 or **/24**	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
First address **192.168.10**.1 or **/24**	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
Last address **192.168.10**.254 or **/24**	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
Broadcast address **192.168.10**.255 or **/24**	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

11.2 IPv4 UNICAST, BROADCAST, AND MULTICAST

- **UNICAST**
  - Unicast transmission is sending a packet to one destination IP address.
  - For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253

- **BROADCAST**
  - Broadcast transmission is sending a packet to all other destination IP addresses.
  - For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts

- **MULTICAST**
  - Multicast transmission is sending a packet to a multicast address group.
  - For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5

11.3 TYPES OF IPV4 ADDRESSES

- **PUBLIC AND PRIVATE IPv4 ADDRESSES**
  - As defined in in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.
  - Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
  - Private IPv4 addresses are not unique and can be used internally within any network. Network Address and Prefix 10.0.0.0/8 172.16.0.0/12 RFC 1918 Private Address Range 10.0.0.0 - 10.255.255.255 172.16.0.0 - 172.31.255.255 192.168.0.0/16
  - However, private addresses are not globally routable

- **ROUTING TO THE INTERNET**
  - Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
  - NAT is typically enabled on the edge router connecting to the internet.
  - It translates the internal private address to a public global IP address

- **SPECIAL USE OF IPv4 ADDRESSES**
  - **Loopback addresses**
    - 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
    - Commonly identified as only 127.0.0.1
    - Used on a host to test if TCP/IP is operational.
  - **Link-Local addresses**
    - 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
    - Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
    - Used by Windows DHCP clients to self-configure when no DHCP servers are available.

- **LEGACY CLASSFUL ADDRESSING**
  - RFC 790 (1981) allocated IPv4 addresses in classes
    - Class A (0.0.0.0/8 to 127.0.0.0/8)
    - Class B (128.0.0.0 /16 – 191.255.0.0 /16)
    - Class C (192.0.0.0 /24 – 223.255.255.0 /24)
    - Class D (224.0.0.0 to 239.0.0.0)
    - Class E (240.0.0.0 – 255.0.0.0)
    - Classful addressing wasted many IPv4 addresses
    - Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

- **ASSIGNMENT OF IP ADDRESSES**
  - The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).
  - RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.

11.4 NETWORK SEGMENTATION
- **BROADCAST DOMAINS AND SEGMENTATION**
  - Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
  - Switches propagate broadcasts out all interfaces except the interface on which it was received.
  - The only device that stops broadcasts is a router.
  - Routers do not propagate broadcasts.
  - Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain

- **PROBLEMS WITH LARGE BROADCAST DOMAINS**
  - A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
  - The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
  - Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.4
  - Broadcasts are only propagated within the smaller broadcast domains.

- **REASONS FOR SEGMENTING NETWORKS**
  - Subnetting reduces overall network traffic and improves network performance.
  - It can be used to implement security policies between subnets.
  - Subnetting reduces the number of devices affected by abnormal broadcast traffic.
  - Subnets are used for a variety of reasons including by:
    - Location
    - Group or Function
    - Device Type

11.5 SUBNET AN IPv4 NETWORK
- **SUBNET OF AN OCTET BOUNDARY**
  - Networks are most easily subnetted at the octet boundary of /8, /16, and /24.
  - Notice that using longer prefix lengths decreases the number of hosts per subnet.
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Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	nnnnnnnn.hhhhhhhh.hhhhhhhh.hhhhhhhh 11111111.00000000.00000000.00000000	16,777,214
/16	255.255.0.0	nnnnnnnn.nnnnnnnn.hhhhhhhh.hhhhhhhh 11111111.11111111.00000000.00000000	65,534
/24	255.255.255.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111111.00000000	254

- **SUBNET WITHIN AN OCTET BOUNDARY**

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Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nhhhhhhh 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnhhhhhh 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnhhhhh 11111111.11111111.11111111.11100000	8	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnhhhh 11111111.11111111.11111111.11110000	16	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 11111111.11111111.11111111.11111000	32	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnhh 11111111.11111111.11111111.11111100	64	2

11.6 SUBNET A SLASH 16 AND A SLASH 8 PREFIX

- **CREATE SUBNETS WITH A SLASH 16 PREFIX**
  - The table highlights all the possible scenarios for subnetting a /16 prefix.

- **CREATE 100 SUBNETS WITH A SLASH 16 PREFIX**
  - Consider a large enterprise that requires at least 100 subnets and has chosen the private address 172.16.0.0/16 as its internal network address.
    - The figure displays the number of subnets that can be created when borrowing bits from the third octet and the fourth octet.
    - Notice there are now up to 14 host bits that can be borrowed (i.e., last two bits cannot be borrowed).\
  - To satisfy the requirement of 100 subnets for the enterprise, 7 bits (i.e., 27 = 128 subnets) would need to be borrowed (for a total of 128 subnets)

- **CREATE 1000 SUBNETS WITH A SLASH 8 PREFIX**
  - Consider a small ISP that requires 1000 subnets for its clients using network address 10.0.0.0/8 which means there are 8 bits in the network portion and 24 host bits available to borrow toward subnetting.
    - The figure displays the number of subnets that can be created when borrowing bits from the second and third.
    - Notice there are now up to 22 host bits that can be borrowed (i.e., last two bits cannot be borrowed).
  - To satisfy the requirement of 1000 subnets for the enterprise, 10 bits (i.e., 210=1024 subnets) would need to be borrowed (for a total of 128 subnets)

11.7 SUBNET TO MET REQUIREMENTS

- **SUBNET PRIVATE VS PUBLIC IPv4 ADDRESS SPACE**
  - Enterprise networks will have an:
    - Intranet - A company's internal network typically using private IPv4 addresses.
    - DMZ –A company's internet facing servers. Devices in the DMZ use public IPv4 addresses.
    - A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.
    - The DMZ devices would have to be configured with public IP addresses

- **MINIMIZE UNUSED HOST IPv4 ADDRESSES AND MAXIMIZE SUBNETS**
  - There are two considerations when planning subnets:
    - The number of host addresses required for each network
    - The number of individual subnets needed

11.8 VLSM

- **IPv4 ADDRESS CONSERVATION**
  - Given the topology, 7 subnets are required (i.e, four LANs and three WAN links) and the largest number of host is in Building D with 28 hosts
    - A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology
  - However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses
    - Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
    - VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet
- **VLSM**
  - The left side displays the traditional subnetting scheme (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to subnet a subnet and divided the last subnet into eight /30 subnets.
  - When using VLSM, always begin by satisfying the host requirements of the largest subnet and continue subnetting until the host requirements of the smallest subnet are satisfied.
  - The resulting topology with VLSM applied.

- **VLSM TOPOLOGY ADDRESS ASSIGNMENT**
  - Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram

11.9 STRUCTURED DESIGN

- **IPv4 NETWORK ADDRESS PLANNING**
  - IP network planning is crucial to develop a scalable solution to an enterprise network.
    - To develop an IPv4 network wide addressing scheme, you need to know how many subnets are needed, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses, and which use public, and many other determining factors.
  - Examine the needs of an organization's network usage and how the subnets will be structured.
    - Perform a network requirement study by looking at the entire network to determining how each area will be segmented.
    - Determine how many subnets are needed and how many hosts per subnet.
    - Determine DHCP address pools and Layer 2 VLAN pools

- **DEVICE ADDRESS ASSIGNMENT**
  - Within a network, there are different types of devices that require addresses:
    - **End user clients** –Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.
    - **Servers and peripherals** –These should have a predictable static IP address.
    - **Servers that are accessible from the internet** –Servers must have a public IPv4 address, most often accessed using NAT.
    - **Intermediary devices** –Devices are assigned addresses for network management, monitoring, and security.
    - **Gateway**–Routers and firewall devices are gateway for the hosts in that network.
  - When developing an IP addressing scheme, it is generally recommended that you have a set pattern of how addresses are allocated to each type of device.

MODULE 12: IPV6 ADDRESSING

12.1 IPv4 ISSUES

- **NEED FOR IPV6**
  - IPv4 is running out of addresses. IPv6 is the successor to IPv4. IPv6 has a much larger 128-bit address space.
  - The development of IPv6 also included fixes for IPv4 limitations and other enhancements.
  - With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT, the time has come to begin the transition to IPv6

- **IPv4 AND IPv6 COEXISTENCE**
  - Both IPv4 and IPv6 will coexist in the near future and the transition will take several years.
  - The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6. These migration techniques can be divided into three categories:
    - Dual stack -The devices run both IPv4 and IPv6 protocol stacks simultaneously.
    - Tunneling–A method of transporting an IPv6 packet over an IPv4 network. The IPv6 packet is encapsulated inside an IPv4 packet.
    - Translation-Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4.

12.2 IPv6 ADDRESS REPRESENTATION

- **IPv6 ADDRESSING FORMATS**
  - IPv6 addresses are 128 bits in length and written in hexadecimal.
  - IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.
  - The preferred format for writing an IPv6 address is x:x:x:x:x:x:x:x, with each "x" consisting of four hexadecimal values.
  - In IPv6, a hextet is the unofficial term used to refer to a segment of 16 bits, or four hexadecimal values.
  - Examples of IPv6 addresses in the preferred format: 2001:0db8:0000:1111:0000:0000:0000:0200 2001:0db8:0000:00a3:abcd:0000:0000:1234

- **RULE 1- OMIT LEADING ZERO**
  - The first rule to help reduce the notation of IPv6 addresses is to omit any leading 0s (zeros).
  - Examples:
    - 01ab can be represented as 1ab
    - 09f0 can be represented as 9f0
    - 0a00 can be represented as a00
    - 00ab can be represented as a

- **RULE 2 - DOUBLE COLON**
  - A double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all zeros.
  - Example:
    - 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1

12.3 IPv6 ADDRESS TYPES

- **UNICAST, MULTICAST, ANYCAST**
  - There are three broad categories of IPv6 addresses:
    - Unicast– Unicast uniquely identifies an interface on an IPv6-enabled device.
    - Multicast– Multicast is used to send a single IPv6 packet to multiple destinations.
    - Anycast–This is any IPv6 unicast address that can be assigned to multiple devices.
  - A packet sent to an anycast address is routed to the nearest device having that address

- **IPv6 PREFIX LENGTH**
  - Prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.
  - The IPv6 prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64.

- **TYPES OF IPv6 UNICAST ADDRESSES**
  - Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:
    - **Global Unicast Address (GUA)** – This is similar to a public IPv4 address. These are globally unique, internet-routable addresses.
    - **Link-local Address (LLA)** - Required for every IPv6-enabled device and used to communicate with other devices on the same local link. LLAs are not routable and are confined to a single link.

- **A NOTE ABOUT THE UNIQUE LOCAL ADDRESS**
  - The IPv6 unique local addresses (range fc00::/7 to fdff::/7) have some similarity to RFC 1918 private addresses for IPv4, but there are significant differences:
    - Unique local addresses are used for local addressing within a site or between a limited number of sites.
    - Unique local addresses can be used for devices that will never need to access another network.
    - Unique local addresses are not globally routed or translated to a global IPv6 address

- **IPv6 GUA**
  - IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet.
    - Currently, only GUAs with the first three bits of 001 or 2000::/3 are being assigned.
    - Currently available GUAs begins with a decimal 2 or a 3 (This is only 1/8th of the total available IPv6 address space)

- **IPv6 GUA STRUCTURE**
  - **Global Routing Prefix:**
    - The global routing prefix is the prefix, or network, portion of the address that is assigned by the provider, such as an ISP, to a customer or site. The global routing prefix will vary depending on ISP policies.
  - **Subnet ID:**
    - The Subnet ID field is the area between the Global Routing Prefix and the Interface ID. The Subnet ID is used by an organization to identify subnets within its site.
  - **Interface ID:**
    - The IPv6 interface ID is equivalent to the host portion of an IPv4 address. It is strongly recommended that in most cases /64 subnets should be used, which creates a 64-bit interface ID.

- **IPv6 LLA**
  - An IPv6 link-local address (LLA) enables a device to communicate with other IPv6 enabled devices on the same link and only on that link (subnet).
    - Packets with a source or destination LLA cannot be routed.
    - Every IPv6-enabled network interface must have an LLA.
    - If an LLA is not configured manually on an interface, the device will automatically create one.
    - IPv6 LLAs are in the fe80::/10 range

12.4 GUA AND LLA STATIC CONFIGURATION

- **STATIC GUA CONFIGURATION ON A ROUTER**
  - Most IPv6 configuration and verification commands in the Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of ipv6 in place of ip within the commands.
    - The command to configure an IPv6 GUA on an interface is: ipv6 address ipv6 address/prefix-length.
    - The example shows commands to configure a GUA on the G0/0/0 interface on R1:

- **STATIC GUA CONFIURATION ON A WINDOWS HOST**

- ○ Manually configuring the IPv6 address on a host is similar to configuring an IPv4 address.
- ○ The GUA or LLA of the router interface can be used as the default gateway. Best practice is to use the LLA.

- **STATIC GUA CONFIGURATION OF A LINK-LOCAL UNICAST ADDRESS**
  - ○ Configuring the LLA manually lets you create an address that is recognizable and easier to remember.
    - ▪ LLAs can be configured manually using the ipv6 address ipv6-link-local-address link-local command.
    - ▪ The example shows commands to configure a LLA on the G0/0/0 interface on R1

12.5 DYNAMIC ADDRESSING FOR IPv6 GUAs

- **RS AND RA MESSAGES**
  - ○ Devices obtain GUA addresses dynamically through Internet Control Message Protocol version 6 (ICMPv6) messages.
    - ▪ Router Solicitation (RS) messages are sent by host devices to discover IPv6 routers
    - ▪ Router Advertisement (RA) messages are sent by routers to inform hosts on how to obtain an IPv6 GUA and provide useful network information such as:
    - ▪ Network prefix and prefix length
    - ▪ Default gateway address
    - ▪ DNS addresses and domain name
    - ▪ The RA can provide three methods for configuring an IPv6 GUA :
    - ▪ SLAAC • SLAAC with stateless DHCPv6 server
    - ▪ Stateful DHCPv6 (no SLAAC)

- **METHOD 1: SLAAC**
  - ○ SLAAC allows a device to configure a GUA without the services of DHCPv6.
  - ○ Devices obtain the necessary information to configure a GUA from the ICMPv6 RA messages of the local router.
  - ○ The prefix is provided by the RA and the device uses either the EUI-64 or random generation method to create an interface ID

- **METHOD 2: SLAAC AND STATELESS DHCP**
  - ○ An RA can instruct a device to use both SLAAC and stateless DHCPv6. The RA message suggests devices use the following:
    - ▪ SLAAC to create its own IPv6 GUA
    - ▪ The router LLA, which is the RA source IPv6 address, as the default gateway address
    - ▪ A stateless DHCPv6 server to obtain other information such as a DNS server address and a domain name

- **METHOD 3: STATEFUL DHCPv6**
  - ○ An RA can instruct a device to use stateful DHCPv6 only.
  - ○ Stateful DHCPv6 is similar to DHCP for IPv4. A device can automatically receive a GUA, prefix length, and the addresses of DNS servers from a stateful DHCPv6 server.
  - ○ The RA message suggests devices use the following:
    - ▪ The router LLA, which is the RA source IPv6 address, for the default gateway address.
    - ▪ A stateful DHCPv6 server to obtain a GUA, DNS server address, domain name and other necessary information

- **EUI-64 PROCESS VS RANDOMLY GENERATED**
  - ○ When the RA message is either SLAAC or SLAAC with stateless DHCPv6, the client must generate its own interface ID.
  - ○ The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number

- **EUI64 PROCESS**
  - ○ The IEEE defined the Extended Unique Identifier (EUI) or modified EUI-64 process which performs the following:
    - ▪ A 16 bit value of fffe (in hexadecimal) is inserted into the middle of the 48-bit Ethernet MAC address of the client.
    - ▪ The 7thbit of the client MAC address is reversed from binary 0 to 1
    - ▪ EXAMPLE:

48-bit MAC	fc:99:47:75:ce:e0
EUI-64 Interface ID	fe:99:47:ff:fe:75:ce:e0

- **RANDOMLY GENERATED INTERFACE IDs**
  - ○ Depending upon the operating system, a device may use a randomly generated interface ID instead of using the MAC address and the EUI-64 process.
  - ○ Beginning with Windows Vista, Windows uses a randomly generated interface ID instead of one created with EUI-64. C:\> ipconfig Windows IP Configuration Ethernet adapter Local Area Connection:

12.6 DYNAMIC ADDRESSING FOR IPv6 LLAs

- **DYNAMIC LLAs**
  - ○ All IPv6 interfaces must have an IPv6 LLA.
  - ○ Like IPv6 GUAs, LLAs can be configured dynamically.

- The figure shows the LLA is dynamically created using the fe80::/10 prefix and the interface ID using the EUI-64 process, or a randomly generated 64-bit number

- **DYNAMIC LLAs ON WINDOWS**
  - Operating systems, such as Windows, will typically use the same method for both a SLAAC-created GUA and a dynamically assigned LLA.

EUI-64 Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . . . . . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
Link-local IPv6 Address . . . . . . : fe80::fc99:47ff:fe75:cee0
Default Gateway . . . . . . . . . . : fe80::1
C:\>
```

Random 64-bit Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
   Link-local IPv6 Address . . . . . . : fe80::50a5:8a35:a5bb:66e1
   Default Gateway . . . . . . . . . . : fe80::1
C:\>
```

- **DYNAMIC LLAs ON CISCO ROUTERS**
  - Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

- **IPv6 ADDRESS CONFIGURATION**
  - Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

12.7 IPv6 MULTICAST ADDRESSES

- **ASSIGNED IPv6 MULTICAST ADDRESSES**
  - IPv6 multicast addresses have the prefix ff00::/8. There are two types of IPv6 multicast addresses:
    - Well-Known multicast addresses
    - Solicited node multicast addresses

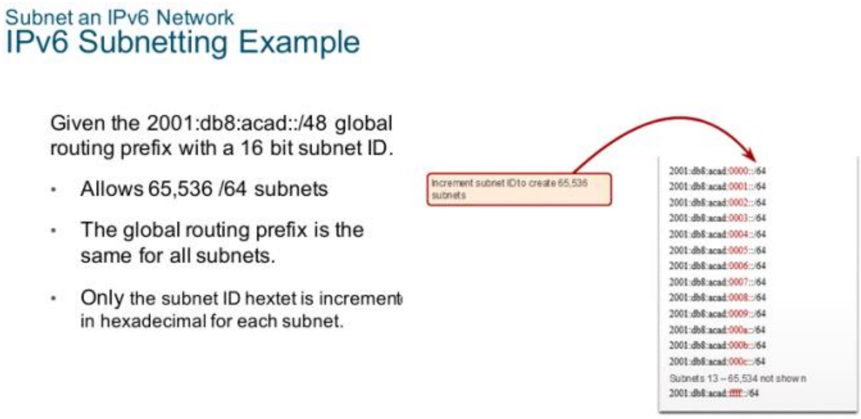
- **WELL-KNOWN IPV6 MULTICAST ADDRESSES**
  - Well-known IPv6 multicast addresses are assigned and are reserved for predefined groups of devices.
  - There are two common IPv6 Assigned multicast groups:
    - ff02::1 All-nodes multicast group- This is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network.
    - ff02::2 All-routers multicast group- This is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an IPv6 router with the ipv6 unicast-routing global configuration command.

- **SOLICITED-NODE IPv6 MULTICAST**
  - A solicited-node multicast address is similar to the all-nodes multicast address.

- A solicited-node multicast address is mapped to a special Ethernet multicast address.
- The Ethernet NIC can filter the frame by examining the destination MAC address without sending it to the IPv6 process to see if the device is the intended target of the IPv6 packet

12.8 SUBNET AN IPV6 NETWORK

- **SUBNET USING THE SUBNET ID**
  - IPv6 was designed with subnetting in mind.
    - A separate subnet ID field in the IPv6 GUA is used to create subnets.
    - The subnet ID field is the area between the Global Routing Prefix and the interface ID
    - EXAMPLE:



- **SUBNET ALLOCATION**
  - The example topology requires five subnets, one for each LAN as well as for the serial link between R1 and R2.
  - The five IPv6 subnets were allocated, with the subnet ID field 0001 through 0005. Each /64 subnet will provide more addresses than will ever be needed

- **ROUTER CONFIGURE WITH IPv6 SUBNET**

The example shows that each of the router interfaces on R1 has been configured to be on a different IPv6 subnet.

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```